

data-model

Data Model — Knowledge Areas & Bidirectional Deep Linking

Feature: specs/002-knowledge-areas-deep-linking | **Date:** 2026-09-01

Normative dependency: the knowledge-model contract in the workshop's documentation is the authority on the ported hierarchy. This document defines the entities this feature adds and how they attach to what already exists; where the two describe the same thing, the contract wins (FR-005).

0. What already exists and is not redefined here

Restated only to fix the boundary — none of this is changed by this feature.

- **Passage** — the smallest addressable unit of source content. 1,101 of them. Carries a minted 26-character identifier, a kind, a chapter scope, an ordering key that is also its start time for transcript segments, a source reference with path and line range, its text, its immutable machine text, a provenance marking, a confidence value, an uncertainty flag and reason, a redaction flag, a change-detection hash that is explicitly **not** identity, a mint time and an ingest run.
- **Chapter** and **Section** — chapters are identified by a zero-padded slug; sections are **derived**, never authored, and carry their own derivation rule and cross-check.
- **Cross-reference** — an existing passage-to-passage relation with a relation type, an origin, a score and a depth.

- **Word timing sidecar** — per chapter, 15,610 records for the chapter that exists, each a word, a start, an end and a probability. Derived; not part of the registry.

1. Identity — one rule, five new kinds

Every entity below mints its identifier through **the same minter** the passage registry uses (D-KG-2). The rule is stated once and applies to all of them:

An identifier is minted once, at creation, and persisted. It is never derived from the entity's text, its title, its slug, its ordinal, its position in a file, or a hash of any of these. It never changes. Two entities never share one.

What this forbids, concretely, because these are the forms that look reasonable:

- an area identified by its slug — breaks when the area is renamed, which is a normal editorial act;
- a term identified by its canonical form — breaks when the canonical form is corrected, which is the single most likely edit a term will ever receive;
- a question identified by an area code plus an ordinal, which is how the reference module does it — breaks when a question is inserted, deleted or reordered;
- a mention identified by a time — breaks on every re-transcription (D-KG-5);
- a lesson section identified by its heading text — breaks on every wording fix.

Each of these is a *content-derived* or *positional* identifier, and the platform's identity contract already forbids both, with paired mutation proofs demonstrating that it catches them.

2. Entities

2.1 Area

The workshop's realisation of the reference module's *module* level (D1).

Field	Meaning	Invariant
identifier	minted, persisted	§1
title	human-readable name	non-empty
summary	one-paragraph statement of what the area covers	non-empty when published
ordinal	display order	unique within the curriculum; not identity
origin	promoted — from the five existing prose areas — or extracted	closed vocabulary
evidence	the mentions that evidence it	at least one when published (FR-008)
evidence certainty	mixed or uncertain-only, derived from the evidencing passages	an area evidenced only by uncertain passages is uncertain-only (FR-012)
lesson sections	ordered	may be empty before materials are authored; not publishable so
question sets	one short, one long	both required when published (FR-034a)
publication review	the recorded review	publication and export are blocked without it (FR-016, FR-048)
chapters	the chapters this area draws evidence from	many-to-many. An area is never owned by a chapter

The many-to-many relation to chapters is load-bearing, not future-proofing for its own sake. With one chapter present, a one-to-many shortcut is indistinguishable in behaviour and cheaper to build. It breaks at the second chapter, when a topic recurs — and it breaks after links exist.

2.2 Term

Field	Meaning	Invariant
identifier	minted, persisted	§1
canonical form	the term's preferred name	non-empty
variants	observed surface forms that resolve to this term	may be empty
areas	the areas this term belongs to	may be several
significance	the recorded measure and its inputs	inspectable ; raw frequency alone is disqualified (FR-013, D-KG-6)
certainty	confident or uncertain	an uncertain term is marked and never presented as confident (FR-011)
evidence	the mentions that evidence it	at least one when published

A term with no evidence does not exist. This is a stronger statement than “is not published”: if its last evidencing passage is redacted, the term is withdrawn from the taxonomy, not merely unlinked (edge case; FR-027).

2.3 Lesson Section

An addressable part of an area's materials, so a link can name a part of an area rather than only the whole. This is what makes the operator's “*proper chapters and part of the chapters*” navigable.

Field	Meaning	Invariant
identifier	minted, persisted	§1
area	owning area	exactly one
heading	the section's title	one of the seven skeleton sections, or a per-topic subsection within one

Field	Meaning	Invariant
ordinal	order within the area	not identity
authorship	authored or assembled	required (FR-017)
citations	the passages its claims rest on	every substantive claim cited or marked editorial (FR-015)

2.4 Question

Field	Meaning	Invariant
identifier	minted, persisted	§1
area	owning area	exactly one
set	short or long	closed
kind	mcq, short, flashcard	ported vocabulary
prompt	the question	non-empty
choices, correct index	multiple choice only	present exactly when kind is mcq
answer	model answer	present for short and flashcard
explanation	why the answer is right	optional
category, difficulty	ported closed vocabularies	difficulty is one of the reference's three values
answer authorship	authored or assembled	assembled answers are marked as such wherever shown (FR-038, SC-020)
citations	the passages that teach the answer	at least one, and every one must resolve, or the question is not served (FR-035, FR-036, D4)
lesson sections	the sections this question assesses	at least one — this is what makes SC-017b measurable

Two invariants distinguish the sets and are checkable rather than felt (D-KG-7):

- a short-set question may cite **one** passage;
- a long-set question **must cite more than one distinct passage**.

This entity is the feature’s one deliberate divergence from the reference. The reference’s equivalent has fields for prompt, choices, answer, explanation, category and difficulty — measured across 785 items — and **no source field of any kind**. The citations field is the difference, and making it mandatory rather than optional is the difference between a guarantee and a claim.

2.5 Mention — the join that carries the feature

Field	Meaning	Invariant
identifier	minted, persisted	§1
subject	the term or area being mentioned	exactly one
passage	the passage it occurs in	the join key. Never a time (FR-033b, D-KG-5)
text span	character offsets within the passage’s text	required
time span	start and end, for media-backed passages	present exactly when the passage is a transcript segment
precision	word or segment	required whenever a time span is present (FR-021, D-KG-1)
timing confidence	the word’s probability, when precision is word	carried, not discarded — 17.4% of words fall below the flag threshold
modality	spoken or on_screen	required on every mention (§2.9, D5). Derived from the passage’s kind, never guessed

Field	Meaning	Invariant
interval bound	the sampling period the time span is accurate to, when the passage is screen_text	carried, not discarded — the on- screen analogue of timing confidence (\$2.9)
origin	how the mention was found	inspectable

Precision is a field, not a convention. A mention resolved by joining the word sidecar has word precision; one resolved only against its enclosing segment has segment. Measured: 99.52% of words join to a segment and 0.48% do not, so both branches are real and the second is not a theoretical fallback. Conflating them is forbidden because the failure is invisible — the link still plays, just in the wrong place.

A mention spanning a segment boundary produces one mention per segment it touches (FR-022). Attaching it to one silently loses half its evidence, and the evidence count is what publication decisions rest on.

2.6 Publication Review

Field	Meaning	Invariant
subject	the area reviewed	exactly one
decision	recorded outcome — including “ <i>nothing to change</i> ”	required. Recording nothing to change is valid; skipping the review is not (FR-016)
reviewer	who	non-empty
time	when	must be newer than the materials it reviews , or it is stale and fails

2.7 Progress

Ported from the reference, whose model already gets the important part right.

Field	Meaning	Invariant
session	whose progress	—
item type, item	what the progress is about	an area, a question set, or a question
status, grade, streak	the learner's state	—

Survives re-ingest (FR-039). The reference deliberately preserves this table when it rebuilds everything else, and that is the behaviour to port: a learner's history is not derived from the content and must not be discarded when the content is rebuilt.

2.8 Locus

Not stored — computed, and returned with every search hit. Named as an entity because the specification requires it to *resolve*, and a thing that must resolve needs a defined shape.

Field	Meaning
containing unit	the area, chapter, lesson section or file that contains the hit
position	a time span for media-backed content; a text span with match offsets otherwise; a file and symbol position for code
precision	inherited from the mention model where the position is a time

The same passage can be reached through more than one containing unit — it may evidence two areas — so the locus is a property of the *hit*, not of the passage.

2.9 On-Screen Text Passage — the `screen_text` kind

Added 2026-09-02 by operator decision (spec §Clarifications, D5). The corpus holds only what was **spoken**; the substance of a screen recording is text that was **shown**. This kind is how that text becomes addressable. It is a new **passage kind**, not a new entity: it takes the existing Passage shape in §0 and fills it, so it inherits the minter, the four-outcome resolver, the redaction flag and the mention model without any of them being touched.

Registry field	How screen_text fills it	Invariant
identifier	minted through the same minter as every other passage	§1. No second minter and no second identifier format
kind	screen_text	joins transcript_segment, doc_section, code, diagram — a fifth value, not a fifth registry
chapter scope	the chapter whose recording it was read from	exactly one
ordering key	the visibility onset — its start time, exactly as a transcript segment's ordering key is its start time	media-backed, so the ordering key is a time
source reference	the recording, plus the sample range the text was observed across	the registry's existing path-plus-range shape; the range counts samples, not lines
text	the recognised text	non-empty. An empty recognition is not a passage — it is a sample that recognised nothing, and is reported as such
immutable machine text	as for every passage	unchanged
provenance marking	ocr, and the engine and settings that produced it	distinct from asr. A reader must be able to tell which engine produced a passage without inferring it from the kind

Registry field	How screen_text fills it	Invariant
confidence	the engine's own confidence for the recognised text	carried, not discarded — the on-screen analogue of N3
uncertainty flag and reason	set when the engine's confidence falls below the recorded threshold	the threshold is recorded, never a literal in code
redaction flag	inherited unchanged	§5 applies to this kind in full
change-detection hash	as for every passage — explicitly not identity	unchanged
mint time, ingest run	as for every passage	unchanged

Its time span is a visibility interval, and the interval has a bound. A transcript segment's time span comes from the recogniser's own alignment. An on-screen passage's comes from a **sampling grid**: the text was first seen at one sample and last seen at another, so the true onset lies within one sampling period of the observed onset. That bound is a real quantity, it is not zero, and it is **carried on the passage and on every mention derived from it** rather than left for a reader to assume away.

Why this does not add a third precision value. The temptation is to mint ocr as a third value of the mention precision field. It is refused: precision is two-valued and every consumer switches on two values (§2.5, contract §3 N1/N2), and a third value changes the wire, the storage and every switch — to express something the field does not mean. precision records **how the time was derived**: from a per-word timing record (word), or from the enclosing unit (segment). An on-screen mention's time is derived from its enclosing visibility interval, so it declares **segment**, and the interval's accuracy travels beside it as the **interval bound** — exactly as timing confidence travels beside word. The vocabulary stays two-valued; the honesty is carried in a field of its own rather than smuggled into an enum.

Honest boundary, measured 2026-09-02. The contract has always been two-valued and word precision is *reachable* — the chapter carries a word sidecar of 15,610 records. It is **not produced in this deployment**: workshop/docs/limits.md §10.1 and the defects row

precision-segment-only in workshop/platform/gates/defects-registry.tsv record that every deep link resolves at segment precision today. This kind therefore joins a model whose second branch is contracted and currently unexercised. Do not read “joins the existing precision model” as “word precision is live”.

2.10 Corroboration Group — what stops a term counting twice

A term said aloud **and** shown on screen at the same moment produces **two** mentions on **two** passages. Both are true and both must be navigable — but they are **one teaching moment**, and the evidence counts that publication rests on (§2.1 *at least one when published*, §2.2, the attached proportion in contract §2.4 A3, and the coverage distribution in contract §4.3 C2) must not silently double because the workshop happened to be recorded with a screen.

Field	Meaning	Invariant
identifier	minted, persisted	§1
subject	the term or area corroborated	exactly one
members	the mentions grouped	at least one ; a lone mention is a group of one, so counting is uniform
modalities	the distinct modalities present across the members	derived, never stored twice
window	the overlap tolerance the grouping used	recorded on the group , so the arithmetic can be audited

- **Grouping is by subject and by overlapping time, within a measured window** — never by text equality alone. Two occurrences of one term twenty minutes apart are two occurrences; deduplicating on text alone collapses a corpus into a vocabulary list.
- **Nothing is deleted.** Both members remain individually retrievable and individually navigable — a learner who wants the moment the term was *shown* gets it. Grouping changes what is **counted**, not what exists.

- **Evidence counting is over groups; mention counting is over mentions**, and the two figures are published side by side so the difference is visible rather than reconciled in someone's head.
- **The window is measured, not assumed.** A term is commonly displayed *before* it is discussed, so a coincidence test tuned to zero lag under-groups, and a window widened until the numbers look tidy over-groups. Its value is settled by measurement (U7, task T130) with a three-valued exit, and task T131 implements the grouping against the value that measurement returns.

A subject evidenced only on screen is marked **on-screen-only** — the modality analogue of uncertain-only (FR-012). It is a different fact from a subject the workshop actually talked about, and a reader who cannot tell them apart will over-read the taxonomy.

3. Relationships — the connectivity matrix, and its storage

FR-033a requires six traversals across four content kinds. Every one is **derived from mentions and citations**; none is a separately stored edge that could drift out of agreement with them.

From	To	Derived through
a moment in the recording	area / lesson / question	the enclosing segment's passage → its mentions → their subjects; and passage → citations → questions
an area, lesson or question	a moment in the recording	citations → passages → time spans, with precision
a transcript passage	area / lesson / question	mentions and citations on that passage
a code passage	a transcript passage	the existing cross-reference graph, plus shared area membership
an area	code passages	mentions whose passage kind is code

From	To	Derived through
any kind	any other, transitively	traversal over the above, cycle-guarded

Every traversal is bidirectional (FR-018), and bidirectionality is a property of the derivation rather than of a second stored edge: a mention read forward gives subject-to-passage and read backward gives passage-to-subject. Storing a reverse edge would be faster and would introduce the possibility of the two disagreeing, which is a class of bug with no symptom.

Cycle safety (FR-026): traversal carries a visited set and a depth bound, and reports truncation rather than silently stopping — the existing cross-reference response already carries a truncation field, and the same discipline applies here.

4. Resolution — one path, four outcomes

Every link, citation, mention and cross-reference resolves through **one** function, the one the platform already uses, with four outcomes that are never collapsed:

Outcome	Meaning
found	the target exists and is readable
redacted	the target exists and its content is suppressed — distinct from absent
not present	the target is not in the registry — a loud dead link
could not determine	the registry could not be read — never collapsed into “not present”

Why the fourth outcome is not optional: collapsing “could not determine” into “not present” makes an unreadable database look like a curriculum that never contained the passage. That is the conflation this repository has found and fixed four separate times, and it is why the platform’s resolver already has this shape.

No fallback of any kind (FR-023, FR-024): no fuzzy text matching, no nearest-neighbour lookup, no prefix matching, no same-content-hash lookup. An unresolvable link fails loudly. A silently re-pointed link is the one outcome the whole identity model exists to prevent.

5. Redaction propagation

FR-027 requires redaction to reach every artifact kind this feature introduces. The propagation targets, enumerated so that none is discovered missing later:

1. **Mentions** on the redacted passage — withdrawn.
2. **Terms** whose last evidence was that passage — the term itself is withdrawn, not merely unlinked.
3. **Areas** whose last evidence was that passage — unpublished, and reported.
4. **Questions** citing it — **withheld**, because an unresolvable citation means the question is not served (FR-036).
5. **Lesson sections** citing it — the claim is marked as having lost its evidence, not silently left standing.
6. **Index entries** — removed; the existing index already excludes redacted rows and the new kinds must inherit that rather than reimplement it.
7. **Exports** — regenerated or withdrawn; an exported document is a copy that outlives the redaction if nothing acts on it.
8. **Stored answers** — any cached answer citing it.

SC-012 measures all eight, with a paired mutation that skips exactly one.

The eight targets are unchanged by the screen_text kind, and that is the point — it is a passage kind, so redacting a screen_text passage withdraws its mentions through target 1 and cascades through the remaining seven by the same rules. What must be proven rather than assumed is that propagation is not written against ASR-derived mentions only; task **T140** asserts it and its paired mutation restricts propagation to the spoken modality.

On-screen text is a wider disclosure surface than the transcript, and the redaction path is not the whole answer to that. A screen recording shows what was never said: window titles, file paths, identifiers, a third party's name in a title bar. The transcript can only carry what somebody spoke; OCR output carries whatever was visible. The boundary obligation this creates is stated as **FR-065** and gated by

G-OCR-11 (task T137), and it applies *before* any OCR-derived text reaches a public artifact — redaction removes what has been recognised as needing removal, and the boundary check is what recognises it.

6. Invariants the model must not be able to violate

Stated as invariants rather than as tests, because a test can be deleted and an invariant enforced in the type cannot be — this is the “an assertion that greps is not a test” rule applied to shape.

- **I1** — no entity carries an identifier derived from its own content or position.
- **I2** — a mention with a time span always carries a precision.
- **I3** — a served question always has at least one resolving citation.
- **I4** — a long-set question always cites more than one distinct passage.
- **I5** — a published area always has at least one evidencing mention and a review newer than its materials.
- **I6** — a published term always has at least one evidencing mention.
- **I7** — no stored relationship is keyed on a time.
- **I8** — every relationship is traversable in both directions.
- **I9** — resolution has exactly four outcomes, and “could not determine” is never one of the other three.
- **I10** — every passage is either attached to an area or explicitly classified as unattached.
- **I11** — the mention `precision` vocabulary has exactly **two** values. A `screen_text` mention declares `segment` and carries its interval bound; it never declares `word`, and no third value exists.
- **I12** — every mention declares its **modality**, and every evidence count that a publication, coverage or attachment figure rests on is taken over **corroboration groups**, never over raw mentions.
- **I13** — no OCR-derived mention is published for a chapter whose OCR accuracy has not been **measured for that chapter**. An unmeasured chapter yields `could-not-determine`, never a default pass.

7. What this model deliberately does not have

- **No area hierarchy.** Areas are flat with an ordering. The reference has a track level above its modules, ported as the curriculum spine, but nesting areas inside areas is not modelled — nothing in the corpus evidences a need for it, and a hierarchy nobody needs is a navigation cost.
- **No authored chapter sections.** Sections stay derived, with their existing rule and cross-check. Authoring them would create a second source of truth about a chapter's structure.
- **No free-text grade.** D3 keeps automatic grading out of scope; a grade field for short answers would invite it back in by accident.
- **No stored reverse edges.** §3 explains why: two representations of one fact can disagree, and the disagreement has no symptom.
- **No third mention precision.** §2.9 explains why: precision records how a time was derived, and an on-screen time is derived from its enclosing interval. Adding an ocr value would change the wire, the storage and every switch in order to express something the field does not mean, while the fact a reader actually needs — how wide the interval is — travels better as a number than as an enum.
- **No deletion of the duplicate occurrence.** §2.10 explains why: the on-screen occurrence is a real moment a learner may want to reach. Only the **count** is deduplicated, and both figures are published.
- **No text-equality deduplication.** Two occurrences of one term at different times are two occurrences. Grouping is by subject **and** overlapping time within a measured window, or it is not grouping — it is a vocabulary list.